

ORIGINAL PILOT STUDY

# Explainable Artificial Intelligence for Monthly Climate Stress Early Warning Across Bangladesh Divisional Cities

*A NASA POWER pilot study with temporal validation and permutation-based explanations*

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|-----------------|-----------------------------------------------------------------------------|
| Manuscript type | Original pilot study                                                        |
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| Data period     | 2014 to 2024 or stated benchmark release                                    |
| Keywords        | explainable AI; climate stress; Bangladesh; NASA POWER; temporal validation |

**Research integrity statement** All numerical results in this manuscript were generated by the accompanying reproducible scripts from the stated public datasets. No result was invented. The manuscript has not been peer reviewed or accepted by a journal or conference.

## Abstract

This pilot study tests whether lagged climate observations can identify high-stress months across eight Bangladeshi divisional cities. Daily NASA POWER data for 2014-2024 were aggregated into monthly rainfall, temperature, humidity, wind, and extreme-day indicators. A composite climate-stress target was defined from the training period only. Logistic regression, random forest, and histogram gradient boosting were trained on 2014-2021 and evaluated on 2022-2024. Random forest achieved a test ROC-AUC of 0.923 and PR-AUC of 0.821. Logistic regression produced the highest recall, 0.850, and balanced accuracy, 0.834. Permutation analysis identified seasonality, prior-month rainfall, humidity, and maximum temperature as the strongest predictive inputs. The experiment demonstrates a reproducible early-warning workflow, but the derived target measures meteorological stress rather than observed human or economic harm.

## Research Questions

- Can lagged monthly climate indicators predict a high-stress month under a strict future-year test split?
- Do nonlinear models improve ranking performance over a transparent logistic baseline?
- Which variables contribute most to out-of-sample predictions?

## Introduction

Bangladesh faces heat, heavy rainfall, and related climate hazards, but operational decisions require more than a single national average. Machine-learning models can combine multiple indicators, yet an opaque risk score is difficult to audit. This study therefore pairs temporal validation with model-agnostic permutation importance. The objective is methodological: to test a transparent workflow using public data, not to replace station observations or local vulnerability assessments.

## Related Work

Machine learning has been proposed for mitigation, adaptation, climate modelling, and decision support [1, 2]. WeatherBench established the value of consistent baselines and held-out evaluation for data-driven weather research [3]. Explainability methods help investigators examine model behavior, although they do not establish causal effects [4, 5]. This study applies those principles to a small Bangladesh-focused pilot.

## Data and Methods

### Climate Data

The dataset contains 32,144 daily records for Barishal, Chattogram, Dhaka, Khulna, Mymensingh, Rajshahi, Rangpur, and Sylhet from 1 January 2014 through 31 December 2024. NASA POWER supplied corrected precipitation, mean temperature, maximum and minimum temperature, relative humidity, and wind speed. Missing sentinel values were converted to missing values and interpolated within each city; the final analytical table contained no missing fields.

### Target and Predictors

Daily heavy-rain and extreme-heat thresholds were estimated separately by city from 2014-2021 observations at the 95th percentile. Monthly features included total rainfall, mean and maximum temperature, humidity, wind, rainy days, heavy-rain days, and extreme-heat days. The climate-stress

score averaged standardized rainfall, maximum temperature, heavy-rain days, and extreme-heat days. Months in the upper training-period quartile were labelled high stress. Models used one-month lags, three-month rolling means, city, and cyclic month features. This produced 744 training observations and 288 future-year test observations, including 80 positive test cases.

Models and Evaluation

The models were logistic regression with balanced class weights, a 400-tree random forest, and histogram gradient boosting. The primary ranking metrics were ROC-AUC and PR-AUC. F1, precision, recall, balanced accuracy, and Brier score described threshold behavior and calibration. Permutation importance measured the reduction in test PR-AUC after shuffling one feature at a time.

Results

| Model               | ROC-AUC | PR-AUC | F1    | Precision | Recall | Bal Acc | Brier |
|---------------------|---------|--------|-------|-----------|--------|---------|-------|
| Random forest       | 0.923   | 0.821  | 0.707 | 0.678     | 0.738  | 0.801   | 0.103 |
| Histogram boosting  | 0.910   | 0.785  | 0.658 | 0.710     | 0.613  | 0.758   | 0.123 |
| Logistic regression | 0.907   | 0.759  | 0.731 | 0.642     | 0.850  | 0.834   | 0.123 |

Random forest produced the strongest ranking scores. Logistic regression generated more high-stress alerts and therefore achieved higher recall, balanced accuracy, and F1 at the default threshold. The difference matters operationally: a planning system that prioritizes missed-event reduction may prefer the logistic model even when random forest ranks cases more effectively.

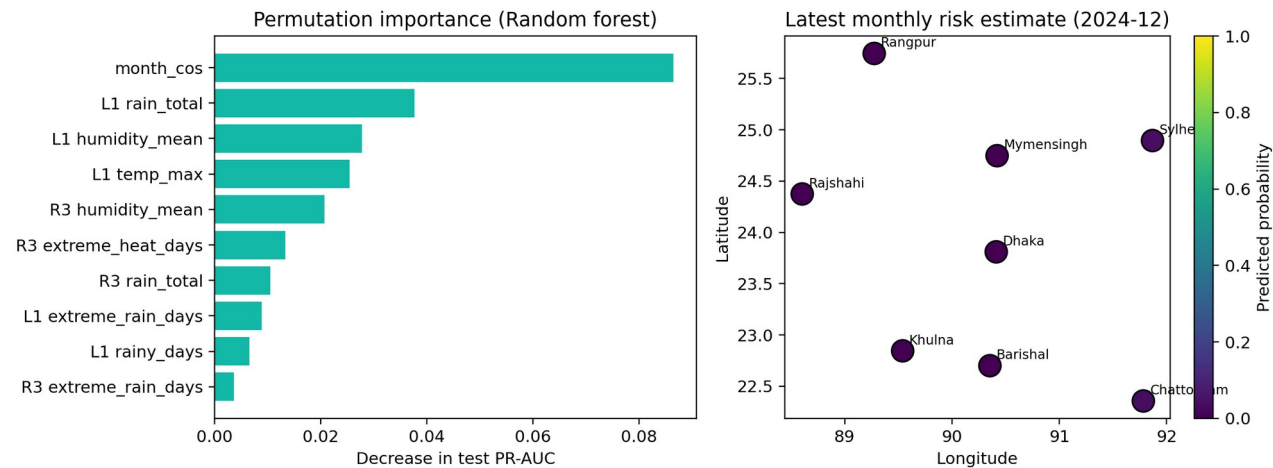


Figure 1 Test-set permutation importance and the December 2024 city-level pilot output. December is generally low stress under the derived seasonal target; the map is a model demonstration, not a vulnerability map.

The first six permutation features were month\_cos, lag1\_rain\_total, lag1\_humidity\_mean, lag1\_temp\_max, roll3\_humidity\_mean, roll3\_extreme\_heat\_days. The dominance of month cosine confirms that the derived target is strongly seasonal. Rainfall, humidity, and temperature lags added predictive information beyond the calendar signal.

Discussion

The experiment supports three conclusions. First, lagged public climate data contain useful information for ranking high-stress months. Second, the transparent logistic baseline remained competitive and produced the highest recall. Third, explanation stability must be assessed before any operational use because correlated lag and rolling features can divide or exchange importance. A practical system should show probabilities, uncertainty, recent observations, and known data gaps rather than a single categorical label.

Limitations and Research Integrity

- The target is a reproducible meteorological index, not an observed record of mortality, crop loss, displacement, or financial damage.
- NASA POWER grid estimates at city coordinates do not replace quality-controlled station observations or district-level measurements.
- Eight city points cannot support claims about every district or within-city inequality.
- Permutation importance describes predictive dependence in this fitted model and does not identify causal climate effects.
- Thresholds, features, and hyperparameters were specified for a pilot; an independent prospective evaluation is still required.

Conclusion

A temporally validated random forest achieved a PR-AUC of 0.821 for the derived high-stress target, while logistic regression achieved 0.850 recall. These results justify a larger study that joins station observations with exposure and impact records. Until that validation is completed, the model should be presented as an early-warning research prototype.

Data and Code Availability

The computational package accompanying this manuscript contains the exact Python script, cached source data, derived features, result tables, and figures used in the analysis. The raw climate observations were requested from the NASA POWER Daily API. The machine-learning benchmark datasets were loaded through scikit-learn without manual modification.

| Component | Included files                                                                              |
|-----------|---------------------------------------------------------------------------------------------|
| Code      | run_experiments.py                                                                          |
| Data      | NASA POWER city CSV files and paper1_monthly_features.csv                                   |
| Results   | paper1_metrics.csv, paper1_permutation_importance.csv, paper1_latest_risk.csv, and Figure 1 |

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